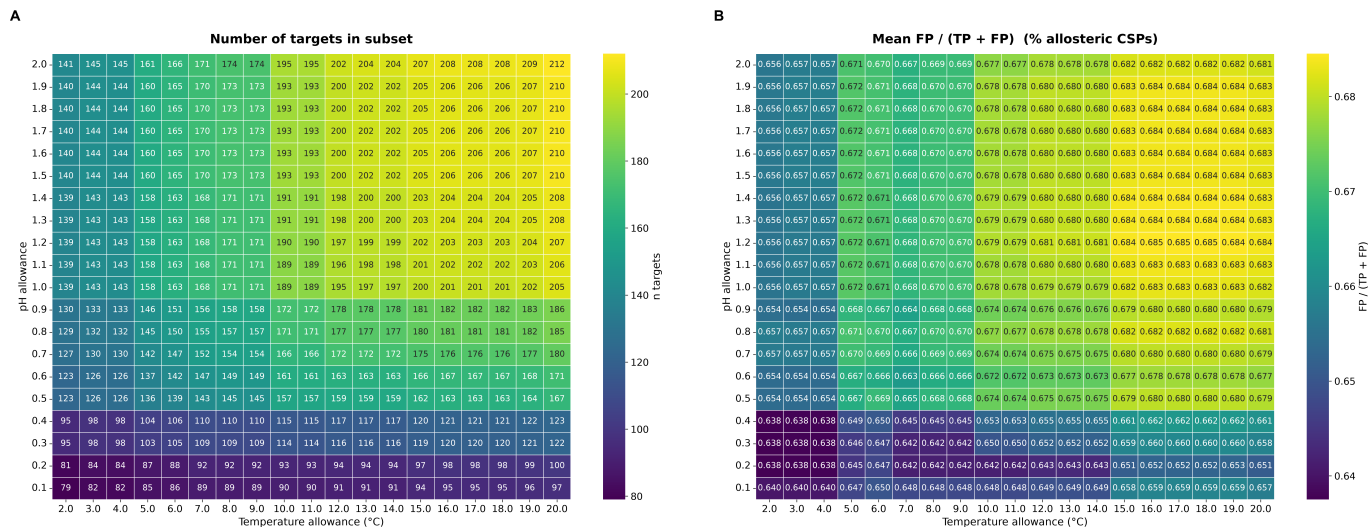
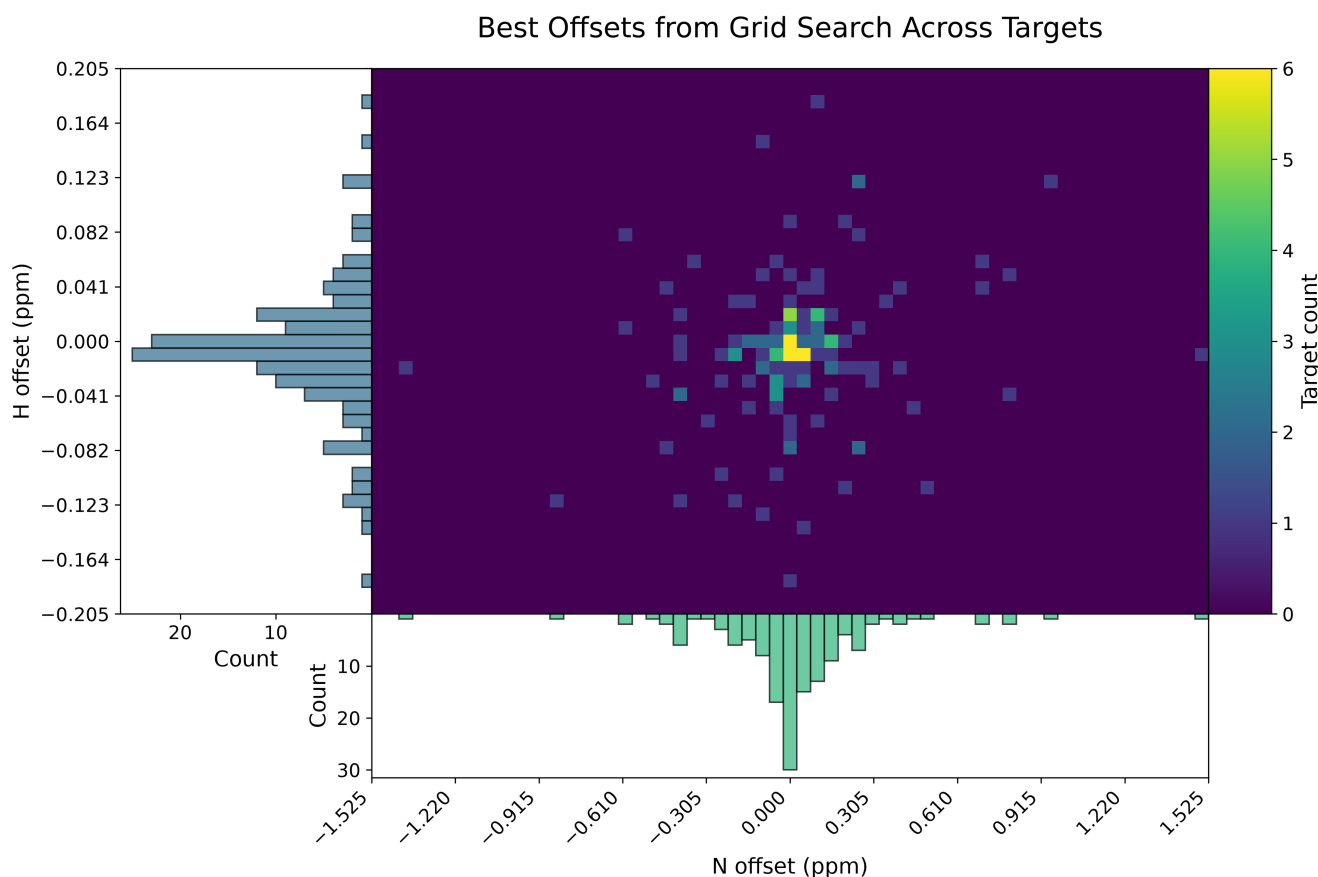




SI Fig. S16. Buffer pH/temperature threshold sweep



SI Fig. S17. Ideal  $^1\text{H}/^{15}\text{N}$  chemical-shift offset grid for the  $n = 145$  buffer-similar systems in CSP\_UBQ\_ph0.5\_temp5C.csv. For each target, a grid search selects the H/N referencing offsets that maximize aligned HN peak counts under the pipeline CSP criterion; the heatmap aggregates those optimal  $(\Delta\delta_{\text{N}}, \Delta\delta_{\text{H}})$  pairs (color = number of targets). Marginal histograms show the one-dimensional distributions of best H (left) and N (bottom) offsets. Both are peaked near zero, indicating that most apo/holo pairs require only small referencing corrections.